Math Basics for Machine Learning
Final Exam

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Instructions

This is the final exam for the Math Basics for Data Science course.

This exam is in two parts:

- PART 1 consists of a quiz similar to the short quizes from our classes. You will need to answer a number of questions, but you do **not** need to provide detailed solutions.
- PART 2 consists of six free-response questions for which you will need to attach detailed solutions, similar to the graded assignments.

You can get 30 points for each of the parts, resulting in 60 points in total for the exam.

You can submit the exam by filling in the corresponding Google form. **You can submit your answers just once.** After you have submitted the form, you should receive a confirmation email. If you have submitted your solutions but did not receive any confirmation, contact me.

You must submit your answers by Monday, November 27, 18:59 Moscow time. ***Late submissions will not be accepted***.

Solutions must be typed in LaTeX. Hand-written solutions, as well as late submissions, will not be accepted.

It is the idea that you complete the exam individually. Do not collaborate, share your solutions with anyone or copy answers of somebody else.

Good luck!

Part 1 (30 points)

The first part of the exam is a quiz. You will need to answer a number of questions, but you do **not** need to provide detailed solutions.

The questions for this part can be found in the submission Google form.

Part 2 (30 points)

The second part of the exam contains six free-response questions for which you will need to attach detailed solutions. You can find problem formulations below.

Always show how you obtain the solution and provide reasonably detailed explanations. Answers stated without any comments won't be accepted. For some tasks, it might be convenient to use Python rather than perform the computations by hand. If you do so, attach your code as well.

1. (4 points) Consider the following three vectors:

$$u = \begin{bmatrix} 4 \\ 5 \\ 0 \end{bmatrix}, \; v = \begin{bmatrix} -1 \\ -4 \\ 3 \end{bmatrix}, \; w = \begin{bmatrix} 10 \\ 7 \\ h+5 \end{bmatrix}$$

where h is some real number.

(a) (2 points) Find all possible values of h for which $\{u, v, w\}$ form a basis for $\mathbb{R}^3$.

Solution: Let's set up a system of equations to check for linear dependence:

$$c_1 u + c_2 v + c_3 w = 0 \quad \text{(where } c_1, c_2, c_3 \text{ are scalars)}$$

$$c_1 \begin{bmatrix} 4 \\ 5 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ -4 \\ 3 \end{bmatrix} + c_3 \begin{bmatrix} 10 \\ 7 \\ h+5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Using Reduced row echelon form to find h we have:

$$\left[\begin{array}{ccc|c} 4 & -1 & 10 & 0 \\ 5 & -4 & 7 & 0 \\ 0 & 3 & h+5 & 0 \end{array}\right]$$

Divide row 1 by 4: $R_1 = \frac{R_1}{4}$.

$$\left[\begin{array}{ccc|c} 1 & -\frac{1}{4} & \frac{5}{2} & 0 \\ 5 & -4 & 7 & 0 \\ 0 & 3 & h+5 & 0 \end{array}\right]$$

Substract row 1 multiplied by 5 from row 2: $R_2 = R_2 - 5R_1$.

$$\left[\begin{array}{ccc|c} 1 & -\frac{1}{4} & \frac{5}{2} & 0 \\ 5 & -\frac{11}{4} & -\frac{11}{2} & 0 \\ 0 & 3 & h+5 & 0 \end{array}\right]$$

Multiply row 2 by $-\frac{4}{11}$: $R_2 = -\frac{4R_2}{11}$.

$$\left[\begin{array}{ccc|c} 1 & -\frac{1}{4} & \frac{5}{2} & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 3 & h+5 & 0 \end{array}\right]$$

Add row 2 multiplied by $\frac{1}{4}$ to row 1:$R_1 = R_1 + \frac{R_2}{4}$.

$$\left[\begin{array}{ccc|c} 1 & 0 & 3 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 3 & h+5 & 0 \end{array}\right]$$

Substract row 2 multiplied by 3 from row 3: $R_3 = R_3 - 3R_2$.

$$\left[\begin{array}{ccc|c} 1 & 0 & 3 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & h-1 & 0 \end{array}\right]$$

From this we can say there is a non-trival solution if $h - 1 = 0$ (that is $h = 1$), and the value of h which makes the vectors linearly dependent is 1 because this will cause c_3 to be a free variable.
However, the only way for this system to have a unique solution (where $c_1 = c_2 = c_3 = 0$) is if $h \neq 1$. If $h = 1$, there are infinitely many solutions, and the set $\{u, v, w\}$ would not be linearly independent.
Therefore, the set $\{u, v, w\}$ forms a basis for $\mathbb{R}^3$ if and only if $h \neq 1$.

(b) (2 points) Pick **one** value of h for which $\{u, v, w\}$ form a basis for $\mathbb{R}^3$ and find the coordinates of the vector $x = (1, 2, 3)$ in that basis.

Solution: Using $h = 2$, to find the coordinates of the vector $x = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ in this basis. We will express x as a linear combination of u, v, and w:

$$c_1 u + c_2 v + c_3 w = x$$

Substitute the values:

$$c_1 \begin{bmatrix} 4 \\ 5 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ -4 \\ 3 \end{bmatrix} + c_3 \begin{bmatrix} 10 \\ 7 \\ 7 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

Now, solve for c_1, c_2, and c_3. This gives us a system of three equations:

$$\begin{aligned} 4c_1 - c_2 + 10c_3 &= 1 \quad .....(1) \\ 5c_1 - 4c_2 + 7c_3 &= 2 \quad .....(2) \\ 3c_2 + 7c_3 &= 3 \quad .....(3) \end{aligned}$$

Selecting equations (1) and (2) and eliminating c_1.

$$\begin{aligned} 4c_1 - c_2 + 10c_3 &= 1 \times 5 \quad \Rightarrow 20c_1 - 5c_2 + 50c_3 = 5 \\ 5c_1 - 4c_2 + 7c_3 &= 2 \times 4 \quad \Rightarrow 20c_1 - 16c_2 + 28c_3 = 2 \end{aligned}$$

it will give: $11c_2 + 22c_3 = -3$ .....(4)
Selecting equations (3) and (4) and eliminating c_2.

$$\begin{aligned} 3c_2 + 7c_3 &= 3 \times 11 \quad \Rightarrow 33c_2 + 77c_3 = 33 \\ 11c_2 + 22c_3 &= -3 \times 3 \quad \Rightarrow 33c_2 + 66c_3 = -9 \end{aligned}$$

This will give: $11c_3 = 42$; Therefore $c_3 = \frac{42}{11}$ .....(5)
Substitute (5) in (3)

$$\begin{aligned} 3c_2 + 7c_3 &= 3 \\ 3c_2 + 7\left(\frac{42}{11}\right) &= -3 \\ 3c_2 = 3 - \frac{294}{11} &= -\frac{261}{11} \\ c_2 &= -\frac{87}{11} \end{aligned}$$

From (1), we have:

$$\begin{aligned} 4c_1 - c_2 + 10c_3 &= 1 \\ 4c_1 - \left(-\frac{87}{11}\right) + 10\left(\frac{42}{11}\right) &= 1 \\ 4c_1 + \frac{507}{11} &= 1 \\ 4c_1 = 1 - \frac{507}{11} &= -\frac{496}{11} \\ c_1 &= \frac{-\frac{496}{11}}{4} \\ c_1 &= -\frac{124}{11} \end{aligned}$$

Therefore the coordinate of x are:
$c_1 = -\frac{124}{11}$; $c_2 = -\frac{87}{11}$; $c_3 = \frac{42}{11}$

2. (3 points) Consider the following linearly independent vectors:

$$u = \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}, \; v = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \; w = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}$$

Apply the Gram-Schmidt process to obtain an *orthonormal* basis for $\mathbb{R}^3$.

Solution: Please find the python link on how I found the orthogonal basis as:

$$\left\{ v_1 = \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}, \; v_2 = \begin{bmatrix} \frac{1}{3} \\ \frac{1}{3} \\ -\frac{2}{3} \end{bmatrix}, \; v_3 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \right\}$$

However, to find the orthonormal basis, we will have to divide each of the vector by its length.

For $v_1 = \sqrt{2^2 + 2^2 + 2^2} = \sqrt{12} = 2\sqrt{3}$

For $v_2 = \sqrt{(\frac{1}{3})^2 + (\frac{1}{3})^2 + (-\frac{2}{3})^2} = \sqrt{\frac{6}{9}} = \frac{\sqrt{6}}{3}$

For $v_3 = \sqrt{1^2 + (-1)^2 + 0^2} = \sqrt{2}$

Dividing each vector with its length, we will have:

$$\left\{ v_1 = \begin{bmatrix} \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{bmatrix}, \ v_2 = \begin{bmatrix} \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \\ -\frac{2}{\sqrt{6}} \end{bmatrix}, \ v_3 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \\ 0 \end{bmatrix} \right\}$$

Therefore, rationalizing the denominator we will have the **orthonormal** basis given as:

$$\left\{ v_1 = \begin{bmatrix} \frac{\sqrt{3}}{3} \\ \frac{\sqrt{3}}{3} \\ \frac{\sqrt{3}}{3} \end{bmatrix}, \ v_2 = \begin{bmatrix} \frac{\sqrt{6}}{6} \\ \frac{\sqrt{6}}{6} \\ -\frac{\sqrt{6}}{3} \end{bmatrix}, \ v_3 = \begin{bmatrix} \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} \\ 0 \end{bmatrix} \right\}$$

3. (3 points) Consider the following matrix:

$$A = \begin{bmatrix} 5 & -6 & -6 \\ -1 & 4 & 2 \\ 3 & -6 & -4 \end{bmatrix}$$

Can it be diagonalized? If so, determine its diagonal form and a basis in which A is diagonal. If not, explain why.

Solution: Please find the python link to my solution. However, the matrix can be diagonalized.

4. (8 points) Let $a_1, ..., a_n$ be some real numbers. Consider the following function:

$$f(x) = \prod_{i=1}^{n} [x^{a_i} e^{-x}], \ x > 0$$

Find the value of $x > 0$ that maximizes $f(x)$.

Solution: To find the value of x that maximizes the function $f(x) = \prod_{i=1}^{n}(x^{a_i} e^{-x})$ for $x > 0$, we can take the natural logarithm of $f(x)$ and then maximize the resulting expression. The natural logarithm simplifies the product into a sum, making it easier to work with.

Let $g(x) = \ln(f(x))$, then:

$$g(x) = \ln\left(\prod_{i=1}^{n}(x^{a_i} e^{-x})\right)$$

$$g(x) = \sum_{i=1}^{n} \ln(x^{a_i} e^{-x})$$

$$g(x) = \sum_{i=1}^{n} (a_i \ln(x) - x)$$

Now, to find the maximum of $g(x)$, we take the derivative of $g(x)$ with respect to x and set it equal to zero:

$$g'(x) = \sum_{i=1}^{n} \left(\frac{a_i}{x} - 1\right) = 0$$

Solving for x:

$$\sum_{i=1}^{n} \frac{a_i}{x} = \sum_{i=1}^{n} 1$$

$$\frac{1}{x} \sum_{i=1}^{n} a_i = n$$

$$x = \frac{1}{n} \sum_{i=1}^{n} a_i$$

Therefore, the value of x that maximizes the function $f(x)$ is $x = \frac{1}{n} \sum_{i=1}^{n} a_i$.

5. (4 points) Find and classify all the critical points of the following function:

$$f(x, y) = 8y - y\sqrt{x-1} + y^3 + 0.5x - 12y^2$$

Note: if the second derivative test is indecisive, you don't need to investigate the correspondent point any further.

Solution:

Finding the partial derivatives:

$$\begin{aligned}\frac{\partial f}{\partial x} &= \frac{\partial}{\partial x}(8y) - \frac{\partial}{\partial x}(y\sqrt{x-1}) + \frac{\partial}{\partial x}\left(y^3\right) + \frac{\partial}{\partial x}(0.5x) - \frac{\partial}{\partial x}\left(12y^2\right) \\ &= 0 - \frac{\partial}{\partial x}(y\sqrt{x-1}) + 0 + \frac{\partial}{\partial x}(0.5x) - 0\end{aligned}$$

Solving for $\frac{\partial}{\partial x}(y\sqrt{x-1})$:

$$\begin{aligned}\frac{\partial}{\partial x}(y\sqrt{x-1}) &= \left(\frac{\partial}{\partial x}(y)\right)\sqrt{x-1} + y\left(\frac{\partial}{\partial x}\left(\sqrt{x-1}\right)\right) \\ &= 0 \cdot \sqrt{x-1} + y\left(\frac{\partial}{\partial x}\left(\sqrt{x-1}\right)\right) = y\left(\frac{\partial}{\partial x}\left(\sqrt{x-1}\right)\right)\end{aligned}$$

Solving for $\frac{\partial}{\partial x}\left(\sqrt{x-1}\right)$:

$$\begin{aligned}\frac{\partial}{\partial x}\left(\sqrt{x-1}\right) &= \frac{1}{2\sqrt{x-1}} \cdot \frac{\partial}{\partial x}(x-1) \\ &= \frac{1}{2\sqrt{x-1}} \cdot 1 = \frac{1}{2\sqrt{x-1}}\end{aligned}$$

$$\begin{aligned}\therefore \quad \frac{\partial f}{\partial x} &= \frac{\partial}{\partial x}(8y) - \frac{\partial}{\partial x}(y\sqrt{x-1}) + \frac{\partial}{\partial x}\left(y^3\right) + \frac{\partial}{\partial x}(0.5x) - \frac{\partial}{\partial x}\left(12y^2\right) \\ &= -\frac{y}{2\sqrt{x-1}} + 0.5\end{aligned}$$

$$\begin{aligned}\frac{\partial}{\partial y}\left(8y - y\sqrt{x-1} + y^3 + 0.5x - 12y^2\right) &= \frac{\partial}{\partial y}(8y) - \frac{\partial}{\partial y}(y\sqrt{x-1}) + \frac{\partial}{\partial y}\left(y^3\right) + \frac{\partial}{\partial y}(0.5x) - \frac{\partial}{\partial y}\left(12y^2\right) \\ &= 8 - \sqrt{x-1} + 3y^2 - 24y\end{aligned}$$

Now, we set both partial derivatives equal to zero and solve for x and y:

1. Setting $\frac{\partial f}{\partial x} = 0$:

$$0.5 - \frac{y}{2\sqrt{x-1}} = 0$$

$$\frac{y}{2\sqrt{x-1}} = 0.5$$

$$y = \sqrt{x-1}$$

2. Setting $\frac{\partial f}{\partial y} = 0$:

$$8 - \sqrt{x-1} + 3y^2 - 24y = 0$$

Substituting $y = \sqrt{x-1}$ into the equation $8 - \sqrt{x-1} + 3y^2 - 24y = 0$ and solve for x and y.
Starting with the given equation:

$$8 - \sqrt{x-1} + 3y^2 - 24y = 0$$

Now substitute $y = \sqrt{x-1}$:

$$8 - \sqrt{x-1} + 3(\sqrt{x-1})^2 - 24\sqrt{x-1} = 0$$

Simplify the expression:

$$8 - \sqrt{x-1} + 3(x-1) - 24\sqrt{x-1} = 0$$

Combine like terms:

$$8 - \sqrt{x-1} + 3x - 3 - 24\sqrt{x-1} = 0$$

Combine constants:

$$5 - \sqrt{x-1} + 3x - 24\sqrt{x-1} = 0$$

Now, let's isolate the square root terms and simplify further:

$$-\sqrt{x-1} - 24\sqrt{x-1} = -5 - 3x$$

Combine like terms on the left side:

$$-25\sqrt{x-1} = -5 - 3x$$

Divide both sides by -25:

$$\sqrt{x-1} = \frac{5+3x}{25}$$

Now, square both sides to eliminate the square root:

$$x - 1 = \left(\frac{5+3x}{25}\right)^2$$

Expand and simplify:

$$x - 1 = \frac{(5+3x)^2}{625}$$

Multiply both sides by 625 to get rid of the fraction:

$$625x - 625 = (5+3x)^2$$

Expand and simplify:

$$625x - 625 = 25 + 30x + 9x^2$$

Move all terms to one side to set the equation to zero:

$$9x^2 - 595x + 650 = 0$$

Solving for x, we will have:

$\Rightarrow 9x^2 - 595x + 650 = 0$
$\Rightarrow 9x^2 - 10x - 585x + 650 = 0$
$\Rightarrow x(9x - 10) - 65(9x - 10) = 0$
$\Rightarrow (9x - 10)(x - 65) = 0$
$\Rightarrow (9x - 10) = 0$ or $(x - 65) = 0$
$\Rightarrow 9x = 10$ or $x = 65$
$\Rightarrow x = \dfrac{10}{9}$ or $x = 65$

Now, that we have found the value of x, we will substitute them back into the equation $y = \sqrt{x-1}$ to find the corresponding values of y.

For $x = \frac{10}{9}$:

$$y = \sqrt{x-1} = \sqrt{\frac{10}{9} - 1} = \sqrt{\frac{1}{9}} = \frac{1}{3}$$

For $x = 65$:

$$y = \sqrt{x-1} = \sqrt{65-1} = \sqrt{64} = 8$$

We have $(x, y) = \left(\frac{10}{9}, \frac{1}{3}\right)$ and $(65, 8)$

To classify the points, we need to find second order partial derivatives.

$$\begin{aligned}\frac{\partial^2}{\partial x^2} &= -\frac{\partial}{\partial x}\left(\frac{y}{2\sqrt{x-1}}\right) + \frac{\partial}{\partial x}(0.5) \quad \text{.....where } \frac{\partial}{\partial x}(0.5) = 0 \\ &= \frac{\frac{\partial}{\partial x}(y)}{2\sqrt{x-1}} + \frac{y\left(\frac{\partial}{\partial x}\left(\frac{1}{\sqrt{x-1}}\right)\right)}{2} \quad \text{.....where } \frac{\partial}{\partial x}(y) = 0\end{aligned}$$

Solving for $\frac{\partial}{\partial x}\left(\frac{1}{\sqrt{x-1}}\right)$:

$$\begin{aligned}\frac{\partial}{\partial x}\left(\frac{1}{\sqrt{x-1}}\right) &= -\frac{1}{2(x-1)^{\frac{3}{2}}} \cdot \frac{\partial}{\partial x}(x-1) \quad \text{.....where } \frac{\partial}{\partial x}(x-1) = 1 \\ &= -\frac{1}{2(x-1)^{\frac{3}{2}}} \\ &= \frac{y\left(-\frac{1}{2(x-1)^{\frac{3}{2}}}\right)}{2} \\ \therefore \quad \frac{\partial^2}{\partial x^2} &= -\left(0 - \frac{y}{4(x-1)^{\frac{3}{2}}} + 0\right) = \frac{y}{4(x-1)^{\frac{3}{2}}} \\ \frac{\partial^2}{\partial y \partial x} &= -\frac{1}{2\sqrt{x-1}}\end{aligned}$$

$$\begin{aligned}\frac{\partial^2}{\partial y^2} &= \frac{\partial}{\partial y}(8) - \frac{\partial}{\partial y}(\sqrt{x-1}) + \frac{\partial}{\partial y}(3y^2) - \frac{\partial}{\partial y}(24y) \\ &= 6y - 24\end{aligned}$$

Lets define $D = f_{xx}f_{yy} - (f_{xy})^2$

If $f_{xx}f_{yy} - (f_{xy})^2 > 0$ and $f_{xx} > 0$, it's a local minimum.
If $f_{xx}f_{yy} - (f_{xy})^2 > 0$ and $f_{xx} < 0$, it's a local maximum.
If $f_{xx}f_{yy} - (f_{xy})^2 < 0$, it's a saddle point.

$$D = f_{xx}f_{yy} - (f_{xy})^2 = \left(\frac{y}{4(x-1)^{\frac{3}{2}}}\right) \cdot (6y-24) - \left(-\frac{1}{2\sqrt{x-1}}\right)^2 = \frac{6y^2 - 24y - \sqrt{x-1}}{4(x-1)^{\frac{3}{2}}}$$

At point $(x, y) = (\frac{10}{9}, \frac{1}{3})$:
Substituting the values in D, we will have;
$D\left(\frac{10}{9}, \frac{1}{3}\right) = -\frac{207}{4}$, which is less than 0. However this point is called a **saddle point**.

At point $(x, y) = (65, 8)$:
Substituting the values in D, we will have;
$D(65, 8) = \frac{23}{256}$, which is greater than 0. Checking further for $f_{xx} = \frac{y}{4(x-1)^{\frac{3}{2}}} = \frac{1}{256}$, which is greater than 0. However this point can be classified as **minimium**

In summary, we can classify the point as:
1. **No relative maximum**
2. **Relative minimum** at point **(x,y)=(65,8)**
3. **Saddle point** at point **(x,y)=($\frac{10}{9}, \frac{1}{3}$)**

6. (8 points) Suppose you want to fit a linear regression model of the form

$$y = w_0 + w_1 \cdot x.$$

You want to do so via **regularized least squares**, i.e., by minimizing the following loss function:

$$\mathcal{L}(x, y, w) = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \lambda \cdot ||w||^2.$$

Here, $\hat{y}_i$ is model's prediction for example i and $||w||^2$ is the l_2 norm of the unknown weights vector $w = (w_0, w_1)$. The effect of adding this extra term to the loss function is that it forces us to chose small values for the unknown coefficients. The larger the value of the hyperparameter λ, the larger is the effect of regularization.

(a) (4 points) You decide to set $\lambda = 0.01$. Compute the gradient of the loss $\mathcal{L}$.

Solution: To compute the gradient of the loss function $\mathcal{L}(x, y, w)$, we need to find the partial derivatives of the loss with respect to each parameter w_i in the weight vector $w = (w_0, w_1)$. Let's denote the i-th component of w as w_i.

The loss function is given by:

$$\mathcal{L}(x, y, w) = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \lambda \cdot ||w||^2.$$

Where $\hat{y}_i$ is the model's prediction for example i, and $||w||^2$ is the l_2 norm of the weight vector.

Let's compute the gradient with respect to w_0 and w_1:

1. Partial derivative with respect to w_0:

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial w_0} &= \sum_{i=1}^{n} \left[\frac{\partial}{\partial w_0} (y_i - (w_0 + w_1 \cdot x_i))^2 \right] + \frac{\partial}{\partial w_0} \lambda w_0^2 \\
\frac{\partial \mathcal{L}}{\partial w_0} &= \sum_{i=1}^{n} [-2(y_i - (w_0 + w_1 \cdot x_i))] + 2\lambda w_0 \\
\frac{\partial \mathcal{L}}{\partial w_0} &= \sum_{i=1}^{n} [-2(y_i - \hat{y}_i)] + 2\lambda w_0 \\
\frac{\partial \mathcal{L}}{\partial w_0} &= -2 \sum_{i=1}^{n} (y_i - \hat{y}_i) + 2\lambda w_0
\end{aligned}$$

2. Partial derivative with respect to w_1:

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial w_1} &= \sum_{i=1}^{n} \left[\frac{\partial}{\partial w_1} (y_i - (w_0 + w_1 \cdot x_i))^2 \right] + \frac{\partial}{\partial w_1} \lambda w_1^2 \\
\frac{\partial \mathcal{L}}{\partial w_1} &= \sum_{i=1}^{n} [-2x_i(y_i - (w_0 + w_1 \cdot x_i))] + 2\lambda w_1 \\
\frac{\partial \mathcal{L}}{\partial w_1} &= \sum_{i=1}^{n} [-2x_i(y_i - \hat{y}_i)] + 2\lambda w_1 \\
\frac{\partial \mathcal{L}}{\partial w_1} &= -2 \sum_{i=1}^{n} x_i(y_i - \hat{y}_i) + 2\lambda w_1
\end{aligned}$$

So, the gradient vector $\nabla\mathcal{L}$ is given by:

$$\nabla\mathcal{L} = \begin{bmatrix} -2\sum_{i=1}^{n}(y_i - \hat{y}_i) + 2\lambda w_0 \\ -2\sum_{i=1}^{n} x_i(y_i - \hat{y}_i) + 2\lambda w_1 \end{bmatrix}.$$

Now, set $\lambda = 0.01$:

$$\nabla\mathcal{L} = \begin{bmatrix} -2\sum_{i=1}^{n}(y_i - \hat{y}_i) + 0.02 w_0 \\ -2\sum_{i=1}^{n} x_i(y_i - \hat{y}_i) + 0.02 w_1 \end{bmatrix}.$$

(b) (4 points) Suppose that you have observed two examples: $x_1 = 1,\ y_1 = 2$ and $x_2 = 2,\ y_2 = -1$. Your initial guess for the unknown weights is $w^0 = (0, 1)$.

Perform one step of the gradient descent algorithm to update parameters' values. Chose learning rate $\eta = 0.1$.

Solution: To perform one step of the gradient descent algorithm, we'll update the weights w using the gradient of the loss function with respect to the weights, multiplied by the learning rate η. The update rule is given by:

$$w^{(t+1)} = w^{(t)} - \eta \cdot \nabla\mathcal{L}$$

Given the examples $x_1 = 1,\ y_1 = 2$ and $x_2 = 2,\ y_2 = -1$, and the initial guess $w^0 = (0, 1)$, the predicted values $\hat{y}_i$ are calculated as follows:

$$\hat{y}_1 = w_0 + w_1 \cdot x_1 = 0 + 1 \cdot 1 = 1$$
$$\hat{y}_2 = w_0 + w_1 \cdot x_2 = 0 + 1 \cdot 2 = 2$$

Now, we can compute the gradient vector $\nabla\mathcal{L}$ using the formulas derived in the previous response:

$$\nabla\mathcal{L} = \begin{bmatrix} -2\sum_{i=1}^{n}(y_i - \hat{y}_i) + 2\lambda w_0 \\ -2\sum_{i=1}^{n} x_i(y_i - \hat{y}_i) + 2\lambda w_1 \end{bmatrix}.$$

Substitute the values and compute:

$$\nabla\mathcal{L} = \begin{bmatrix} -2\left((2-1) + (-1-2)\right) + 2 \cdot 0.01 \cdot 0 \\ -2\left(1 \cdot (2-1) + 2 \cdot (-1-2)\right) + 2 \cdot 0.01 \cdot 1 \end{bmatrix}$$
$$\nabla\mathcal{L} = \begin{bmatrix} 4 \\ 10.02 \end{bmatrix}$$

Now, use the gradient descent update rule with the learning rate $\eta = 0.1$:

$$w^{(1)} = w^{(0)} - \eta \cdot \nabla\mathcal{L}$$

$$w^{(1)} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} - 0.1 \cdot \begin{bmatrix} 4 \\ 10.02 \end{bmatrix}$$

$$w^{(1)} = \begin{bmatrix} 0 - 0.4 \\ 1 - 1.002 \end{bmatrix} = \begin{bmatrix} -0.4 \\ -0.002 \end{bmatrix}$$

After one step of the gradient descent algorithm, the updated weights are $w^{(1)} = (-0.4, -0.002)$